

# **Problem Solving Using Python (CSE1012)**

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01 Matrix operations using Numpy

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# Creating Matrix using Numpy

```
import numpy
```

```
A = numpy.array([[1, 2], [3, 4]])
```

```
print(A)
```

```
[[1 2]
```

```
[3 4]]
```

# Creating Matrix using Numpy

```
import numpy
```

```
A = numpy.array([[1, 2], [3, 4]])
```

```
B = numpy.array([[1, 1], [1, 1]])
```

```
print(A+B)
```

```
[[2 3]
```

```
[4 5]]
```

# Creating Matrix using Numpy

```
import numpy
```

```
C = numpy.zeros((2,3))
```

# Creating Matrix using Numpy

```
import numpy
```

```
C = numpy.zeros((2,3))
```

```
A = numpy.ones((3,5))
```

# Creating Matrix using Numpy

```
import numpy
```

```
A = numpy.arange(4)
```

```
print(A)
```

```
[0 1 2 3]
```

# Creating Matrix using Numpy

```
import numpy
A=numpy.arange(10)
print(A)
[0 1 2 3 4 5 6 7 8 9]
```

```
B=A.reshape(2,5)
print(B)
[[0 1 2 3 4]
 [5 6 7 8 9]]
```



# Accessing Matrix Elements

```
import numpy as np
```

```
A = np.array([[0, 2, 3], [3, 2, 1]])
```

```
print(A)
```

```
[[0 2 3]
```

```
 [3 2 1]]
```

```
print(A[1,2])    #or A[1][2]
```

```
1
```

# Matrix Addition

```
import numpy as np
A = np.array([[1, 2], [1, 1]])
B = np.array([[1, 0], [1, 1]])
C = A + B
print(C)
[[2 2]
 [2 2]]
```

# Matrix Multiplication

```
import numpy as np
A = np.array([[1, 2], [1, 1]])
B = np.array([[1, 0], [1, 1]])
C = A.dot(B)
print(C)
[[3 2]
 [2 1]]
```

# Matrix Multiplication

```
import numpy as np
A = np.array([[1, 2], [1, 1]])
B = np.array([[2], [1]])
C = A.dot(B)
print(C)
[[4]
 [3]]
```

# Matrix Multiplication

```
import numpy as np
A = np.array([[1, 2], [1, 1]])
B = np.array([[2], [1]])
C = B.dot(A)
```

Traceback (most recent call last):

File "<pyshell#47>", line 1, in <module>

B.dot(A)

ValueError: shapes (2,1) and (2,2) not aligned: 1 (dim 1) != 2 (dim 0)

# Matrix Transpose

```
import numpy as np
A = np.array([[1, 2, 3], [3, 2, 1]])
print(A)
[[1 2 3]
 [3 2 1]]
B = A.transpose()
print(B)
[[1 3]
 [2 2]
 [3 1]]
```

# Thank You

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